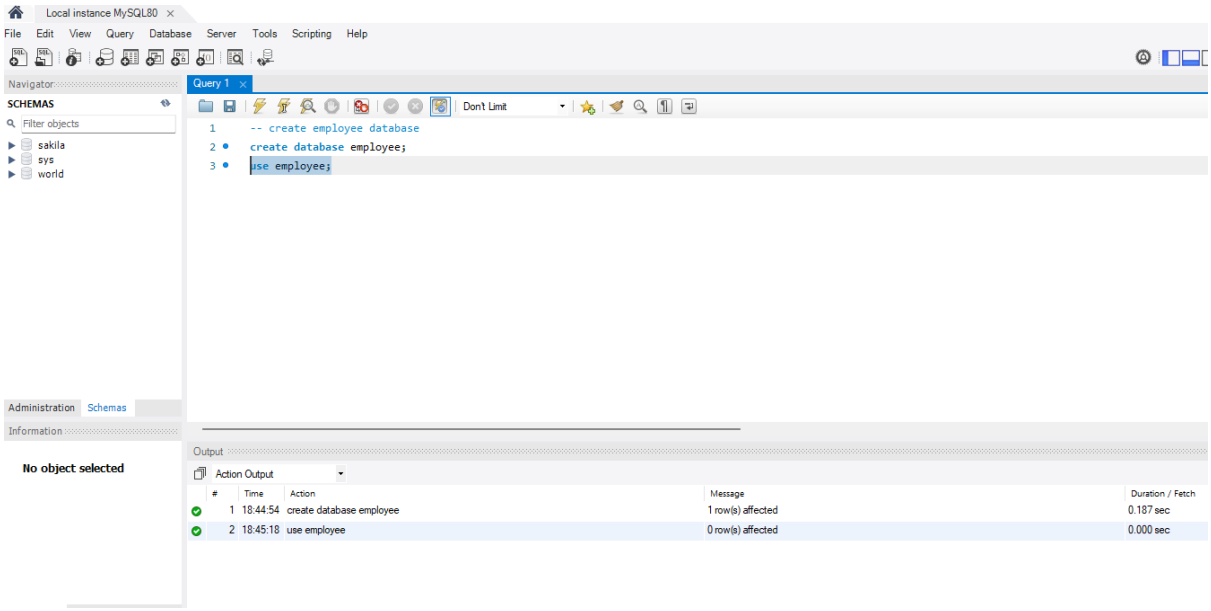
DDL Commands

1. Table Creation (CREATE): Write the SQL statements to create a database named “employee” and the following tables based on the provided schema:
   - ◉ Departments
   - ◉ Location
   - ◉ Employees



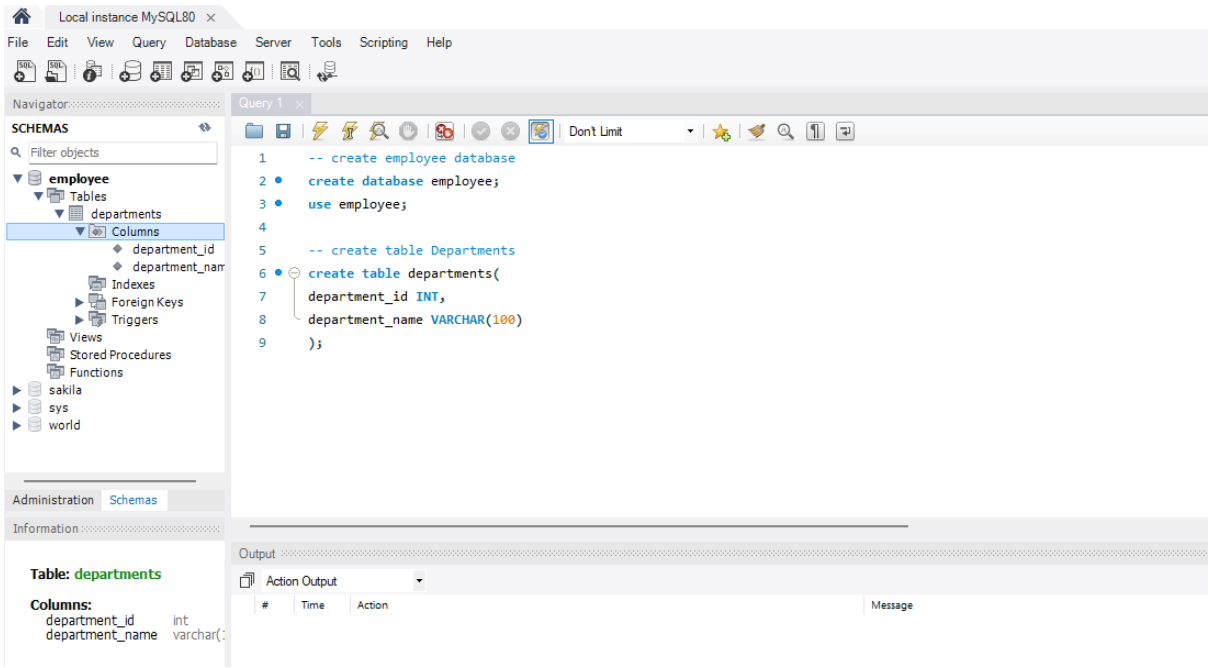


Table Alteration (ALTER):

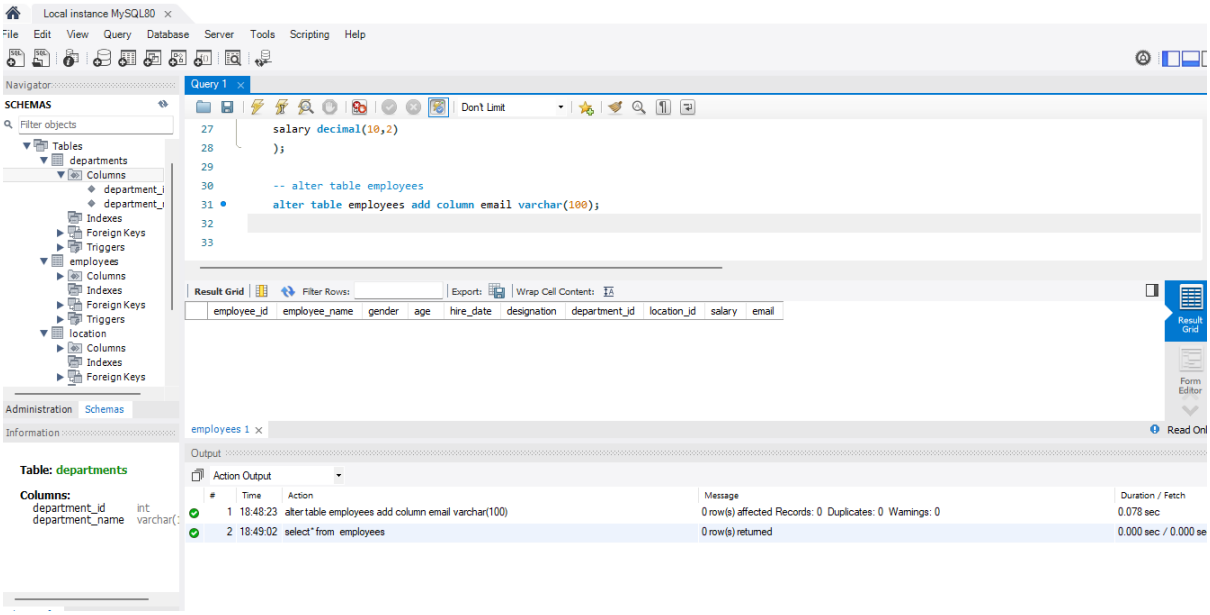
Consider the following scenarios and write the SQL statements to alter the structure of the tables accordingly:

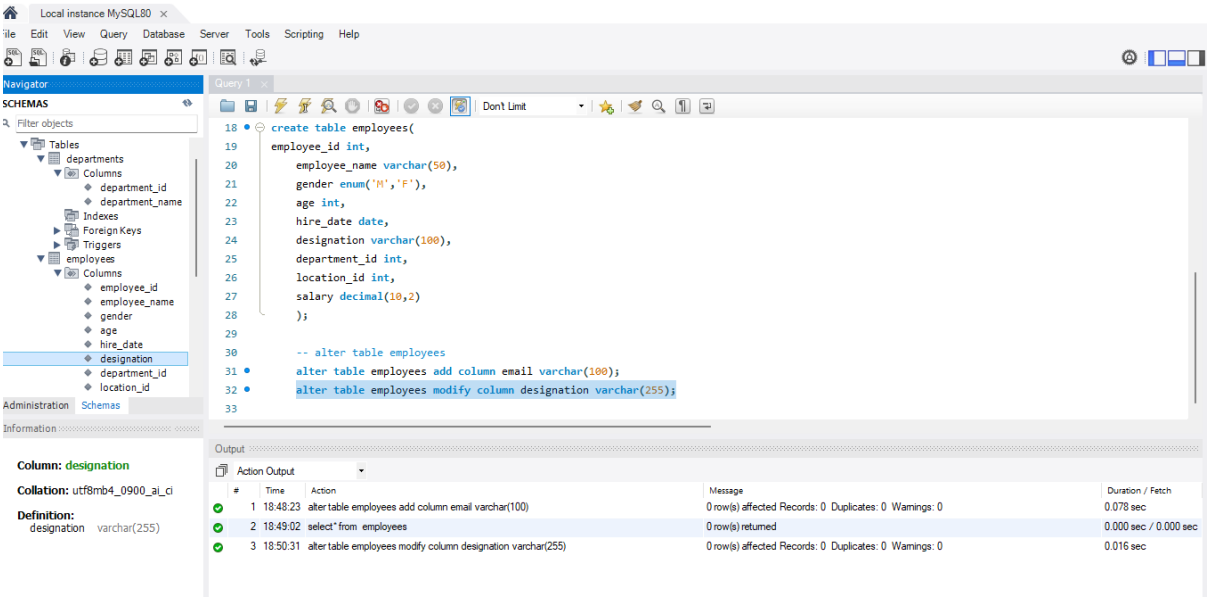
◉ Add a new column named "email" to the Employees table to store employee email addresses.

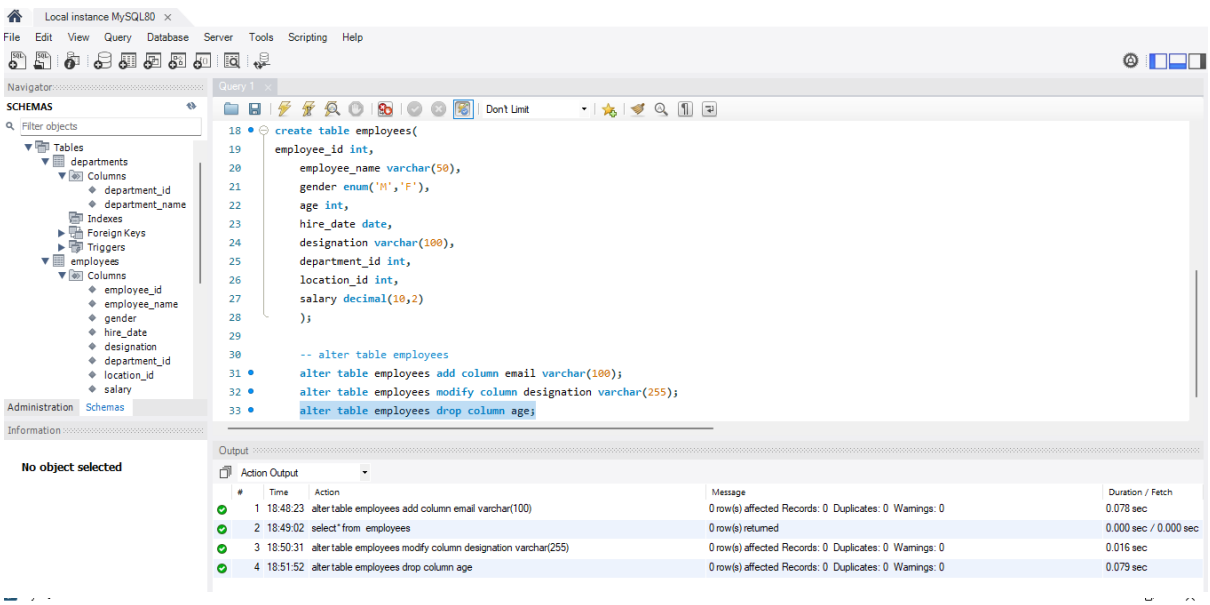
◉ Modify the data type of the "designation" column in the Employees table to support a wider range of values.

◉ Drop the “age” column from the Employees table.

◉ Rename the “hire_date” column to “date_of_joining”.







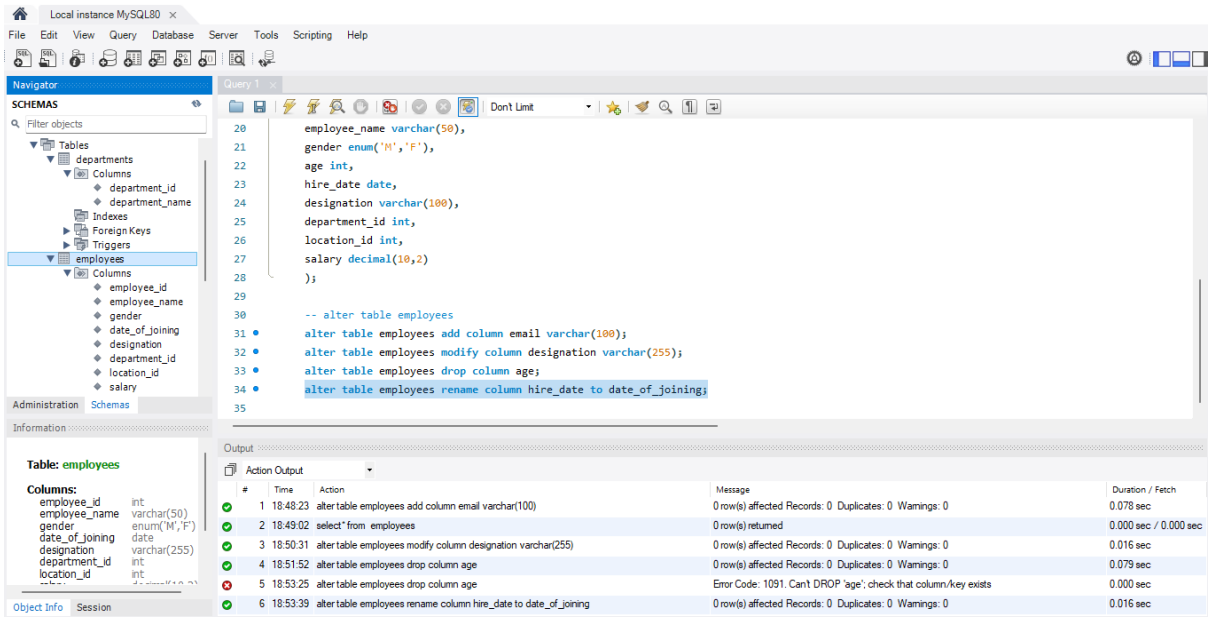
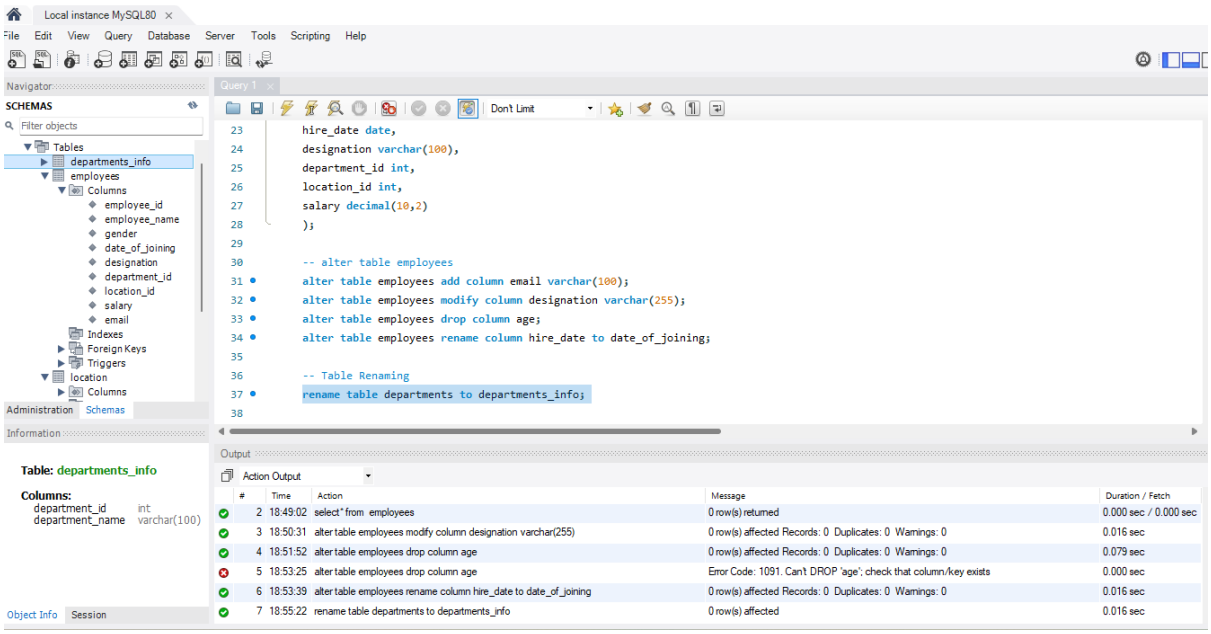


Table Renaming (RENAME)

Rewrite the SQL statements to rename the following tables:

◉ Rename the "Departments" table to "Departments_Info".

◉ Rename the "Location" table to "Locations".


Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
Tables
departments_info
employees
Columns
employee_id
employee_name
gender
date_of_joining
designation
department_id
location_id
salary
email
Indexes
Foreign Keys
Triggers
location
Columns
Administration Schemas
Information
Table: departments_info
Columns:
department_id int
department_name varchar(100)
Object Info Session
Query 1
Don't Limit
23 hire_date date,
24 designation varchar(100),
25 department_id int,
26 location_id int,
27 salary decimal(10,2)
28);
29
30 -- alter table employees
31 alter table employees add column email varchar(100);
32 alter table employees modify column designation varchar(255);
33 alter table employees drop column age;
34 alter table employees rename column hire_date to date_of_joining;
35
36 -- Table Renaming
37 rename table departments to departments_info;
38
Output
Action Output
Time Action Message Duration / Fetch
2 18:49:02 select * from employees 0 row(s) returned 0.000 sec / 0.000 sec
3 18:50:31 alter table employees modify column designation varchar(255) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.016 sec
4 18:51:52 alter table employees drop column age 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.079 sec
5 18:53:25 alter table employees drop column age Error Code: 1091. Can't DROP 'age'; check that column/key exists 0.000 sec
6 18:53:39 alter table employees rename column hire_date to date_of_joining 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.016 sec
7 18:55:22 rename table departments to departments_info 0 row(s) affected 0.016 sec

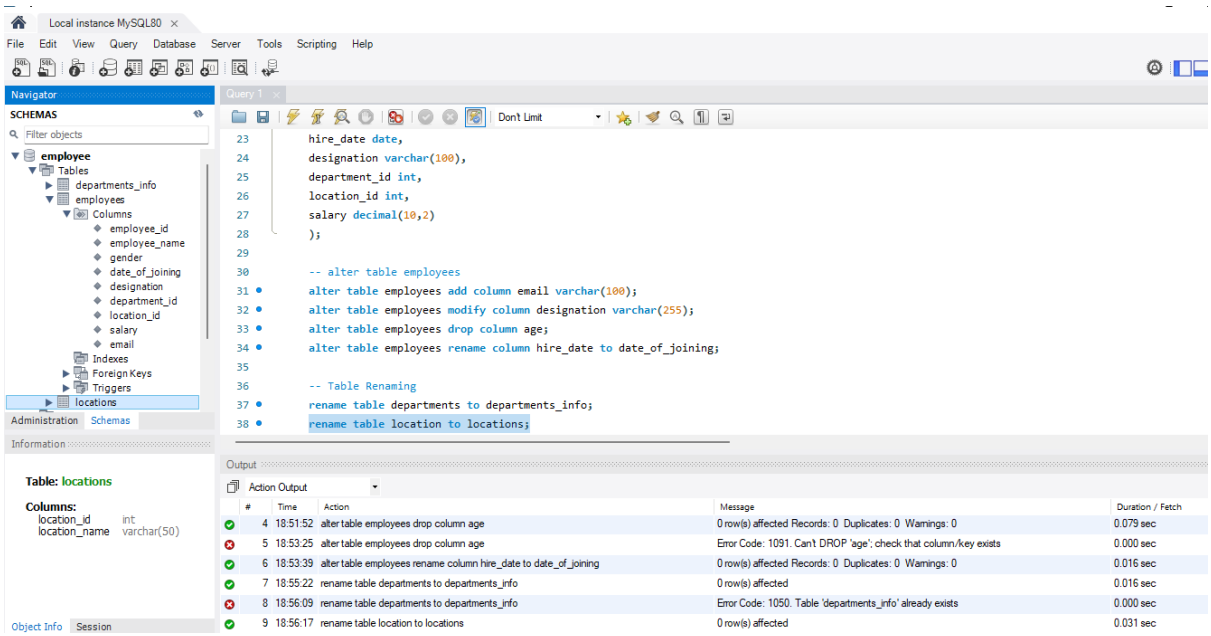
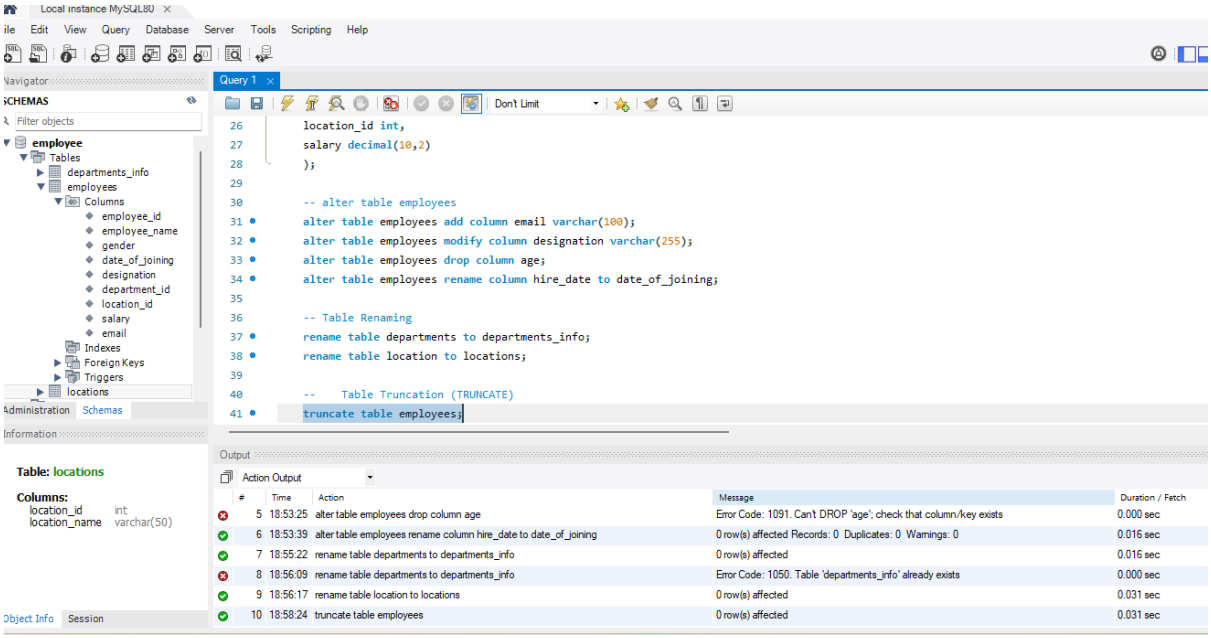

Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
employee
Tables
departments_info
employees
Columns
employee_id
employee_name
gender
date_of_joining
designation
department_id
location_id
salary
email
Indexes
Foreign Keys
Triggers
locations
Administration Schemas
Information
Table: locations
Columns:
location_id int
location_name varchar(50)
Object Info Session
Query 1
Don't Limit
23 hire_date date,
24 designation varchar(100),
25 department_id int,
26 location_id int,
27 salary decimal(10,2)
28);
29
30 -- alter table employees
31 alter table employees add column email varchar(100);
32 alter table employees modify column designation varchar(255);
33 alter table employees drop column age;
34 alter table employees rename column hire_date to date_of_joining;
35
36 -- Table Renaming
37 rename table departments to departments_info;
38 rename table location to locations;
Output
Action Output
Time Action Message Duration / Fetch
4 18:51:52 alter table employees drop column age 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.079 sec
5 18:53:25 alter table employees drop column age Error Code: 1091. Can't DROP 'age'; check that column/key exists 0.000 sec
6 18:53:39 alter table employees rename column hire_date to date_of_joining 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.016 sec
7 18:55:22 rename table departments to departments_info 0 row(s) affected 0.016 sec
8 18:56:09 rename table departments to departments_info Error Code: 1050. Table 'departments_info' already exists 0.000 sec
9 18:56:17 rename table location to locations 0 row(s) affected 0.031 sec

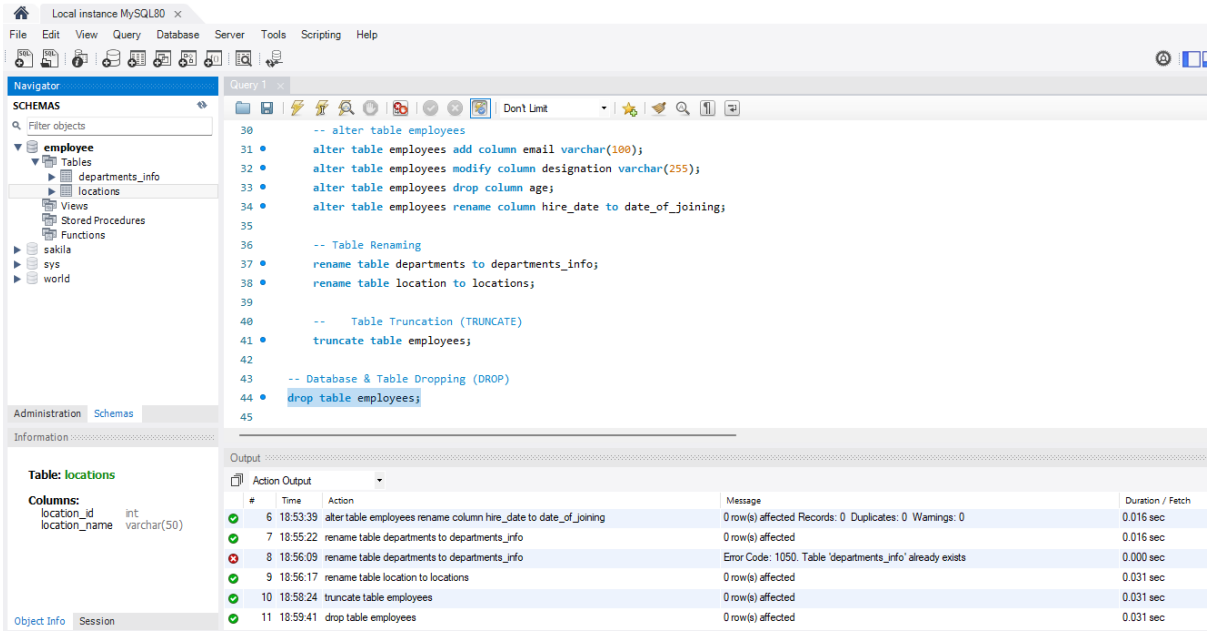
Table Truncation (TRUNCATE):

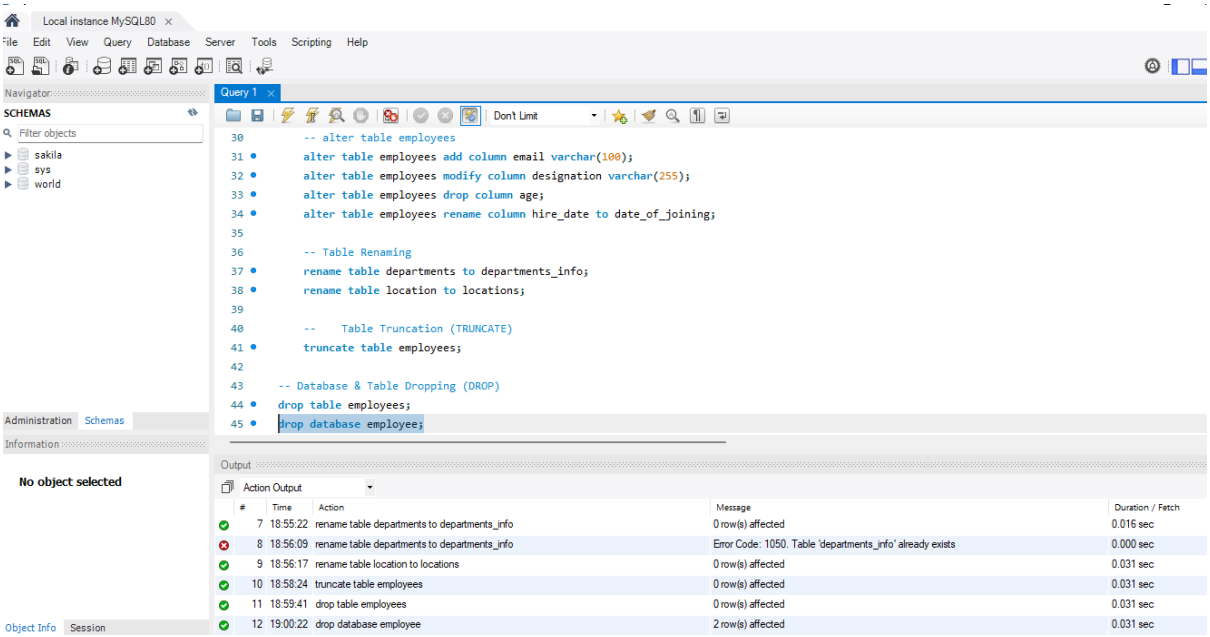
Write an SQL statement to truncate the Employees table.



Database & Table Dropping (DROP)

Write the SQL statements to drop the Employees table and then the “employee” databa

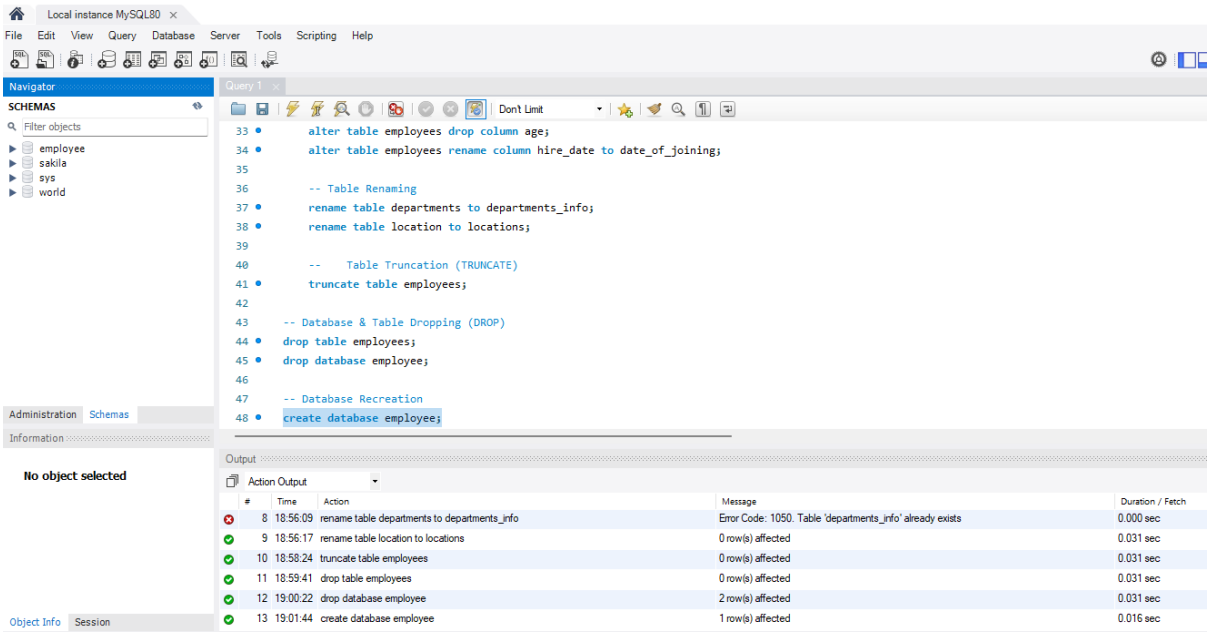


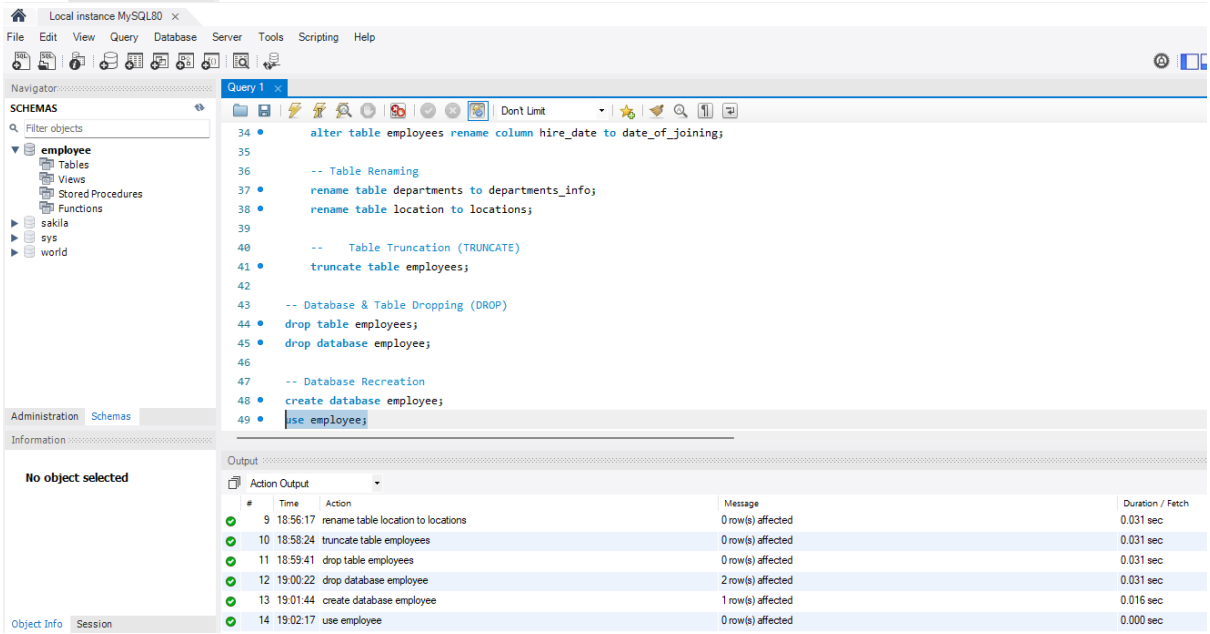


Constraints

1. Database Recreation:

   ◉ Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.

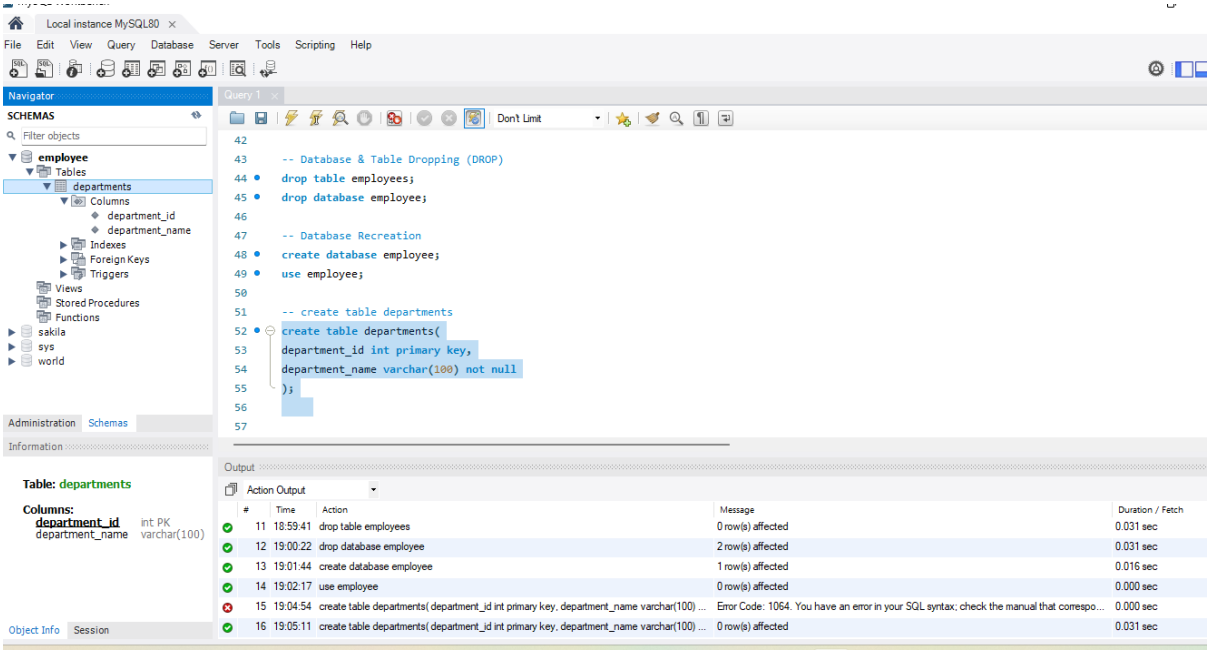




2. Departments Table:

◉ Ensure that the "department_id" uniquely identifies each department.

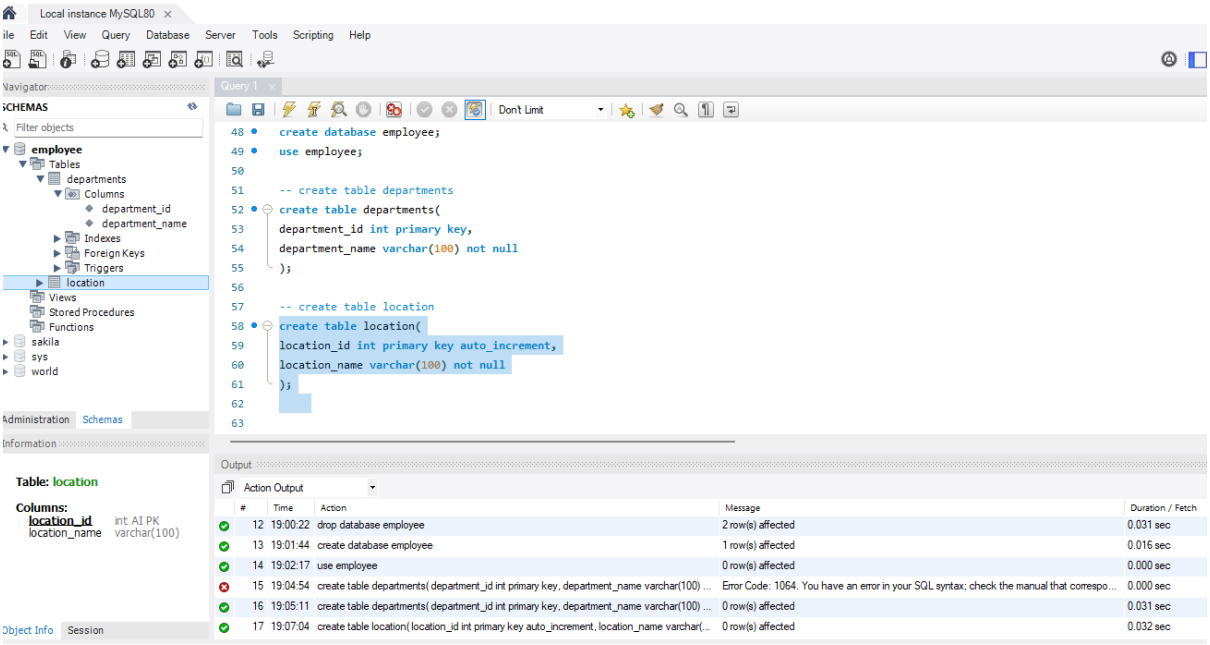
◉ Set up constraints on the "department_name" to avoid duplicate and null entries.



3. Location Table:

◉ Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.

◉ Implement constraints to prevent the insertion of null and duplicate locations.



4. Employees Table:

◉ Guarantee that each employee has a distinct identifier.

◉ Create a restriction to ensure that the employee's name is always provided.

◉ Limit the acceptable values for the "gender" field to only 'M' or 'F'.

◉ Enforce a condition to ensure that the employee's age is 18 or above.

◉ Automatically assign the current date to the "hire_date" field if not specified.

◉ Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

